

C PROGRAMMING

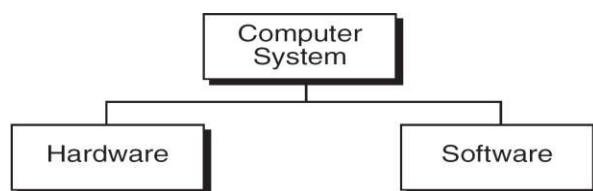


UNIT I

Computer systems:

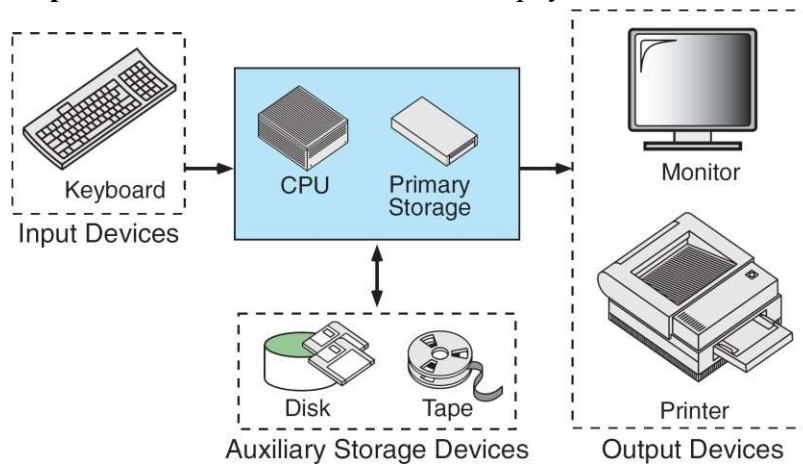
A Computer is an electronic device which performs operations such as accepts data As an input, store the data, manipulate or process the data and produce the results an output.
Main task performed by a computer

- Accept the data
- Process or manipulate the data
- Display or store the result in the form of human understanding
- Store the data, instructions and results.

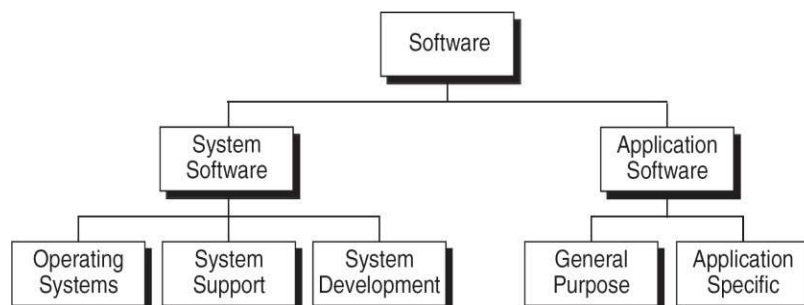


A computer system consists of hardware and software.

Computer hardware is the collection of physical elements that comprise a computer system.



Computer software is a collection of computer programs and related data that provides the instructions for a computer what to do and how to do it. Software refers to one or more computer programs and data held in the storage of the computer for some purpose



Basically computer software is of three main types

System Software: System software is responsible for managing a variety of independent hardware components, so that they can work together. Its purpose is to unburden the application software programmer from the often complex details of the particular computer being used, including such accessories as communications devices, printers, device readers, displays and keyboards, and also to partition the computer's resources such as memory and processor time in a safe and stable manner.

- Device drivers
- Operating systems
- Servers
- Utilities
- Window systems

Programming Software: Programming Software usually provides tools to assist a programmer in writing computer programs, and software using different programming languages in a more convenient way. The tools include:

- Compilers
- Debuggers
- Interpreters
- Linkers
- Text editors

Application Software: Application software is developed to aid in any task that benefits from computation. It is a broad category, and encompasses Software of many kinds, including the internet browser being used to display this page. This category includes:

- Business software
- Computer aided design
- Databases
- Decision making software
- Educational software
- Image editing

Computing Environment:

Computing Environment is a collection of computers / machines, software, and networks that support the processing and exchange of electronic information meant to support various types of computing solutions.

Types of Computing Environments:

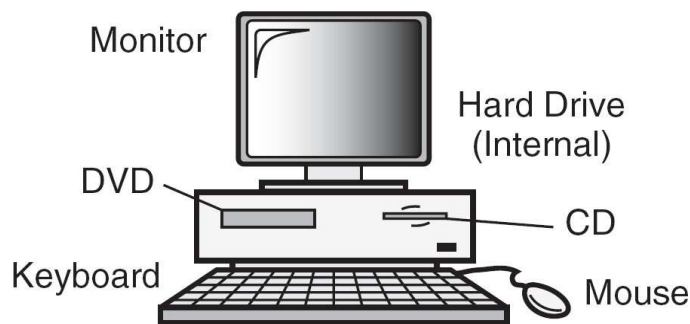
- Personal Computing Environment

- Client Server Environment
- Time sharing Environment
- Distributed Environment

Personal Computing Environment:

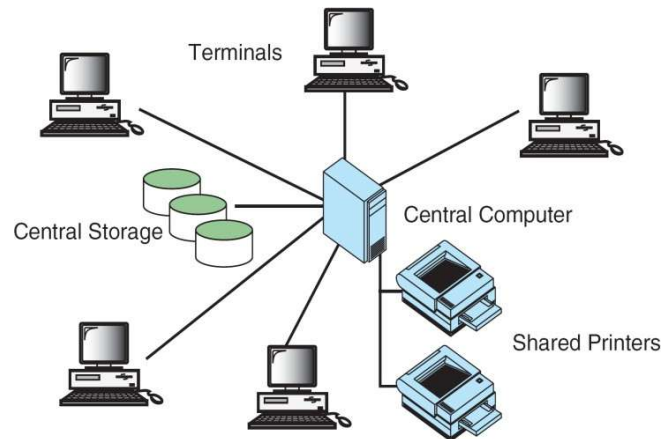
All of the computer hardware components are tied together in our personal computer. A **personal computer (PC)** is a computer whose original sales price, size, and capabilities make it useful for individuals, and intended to be operated directly by an end user, with no intervening computer operator. People generally relate this term with Microsoft's Windows Operating system. Personal computers generally run on Windows, Mac or some version of Linux operating system.

Desktop: Desktop computer is just another version of Personal Computer intended for regular use from a single user. A computer that can be fit on a desk can also be called as desktop.



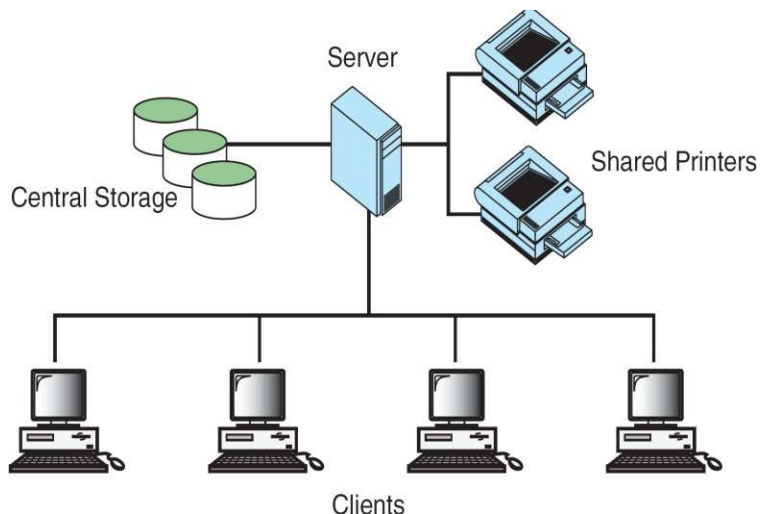
Time-Sharing Environment:

In the time-sharing environment, all computing must be done by the central computer. The central computer manages the shared resources, it manages the shared data and printing. Employees in large companies often work in what is known as time sharing environment. In the time sharing environment, many users are connected to one or more computers. These computers may be mini computers and central mainframes. In this environment the output devices, auxiliary storage devices are shared by all the users.



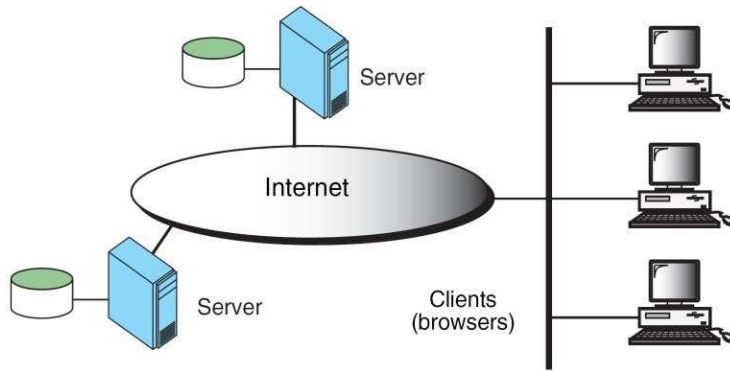
Client/Server Environment

Client/Server computing environment splits the computing function between a central computer and user's computers. The users are given personal computers or work stations so that some of the computation responsibility can be moved from the central computer and assigned to the workstations. In the client/server environment the users micro computers or work stations are called the client. The central computer which may be a powerful micro computer, minicomputer or central mainframe system is known as server.



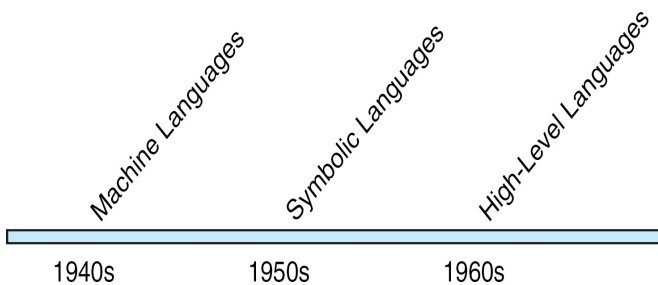
Distributed Computing Environment

A distributed computing environment provides a seamless integration of computing functions between different servers and clients. The internet provides connectivity to different servers throughout the world. This environment provides reliable, scalable and highly available network.



COMPUTER LANGUAGES

In order to communicate with the computer user also needs to have a language that should be understood by the computer. For this purpose, different languages are developed for performing different types of work on the computer. Basically, languages are divided into two categories according to their interpretation.



1. Low Level Languages.

2. High Level Languages.

Low Level Languages

Low level computer languages are machine codes or close to it. Computer cannot understand instructions given in high level languages or in English. It can only understand and execute instructions given in the form of machine language i.e. language of 0 and 1. There are two types of low level languages:

- **Machine Language.**
- **Assembly Language**

Machine Language: It is the lowest and most elementary level of Programming language and was the first type of programming language to be Developed. Machine Language is basically the only language which computer Can understand. In fact, a manufacturer designs a computer to obey just one Language, its machine code, which is represented inside the computer by a String of binary digits (bits) 0 and 1. The symbol 0 stands for the absence of Electric pulse and 1 for the

presence of an electric pulse . Since a computer is Capable of recognizing electric signals, therefore, it understand machine Language.

1		00000000	00000100	0000000000000000
2	01011110	00001100	11000010	0000000000000010
3		11101111	00010110	00000000000000101
4		11101111	10011110	00000000000001011
5	11111000	10101101	11011111	0000000000010010
6		01100010	11011111	0000000000010101
7	11101111	00000010	11111011	0000000000010111
8	11110100	10101101	11011111	0000000000011110
9	00000011	10100010	11011111	0000000000100001
10	11101111	00000010	11111011	0000000000100100
11	01111110	11110100	10101101	
12	11111000	10101110	11000101	0000000000101011
13	00000110	10100010	11111011	0000000000110001
14	11101111	00000010	11111011	0000000000110100
15		01010000	11010100	0000000000111011
16			00000100	0000000000111101

Advantages of Machine Language

- i) It makes fast and efficient use of the computer.
- ii) It requires no translator to translate the code i.e. Directly understood by the computer

Disadvantages of Machine Language:

- i) All operation codes have to be remembered
- iv) These languages are machine dependent i.e. a particular

Machine language can be used on only one type of computer

Assembly Language

It was developed to overcome some of the many inconveniences of machine language. This is another low level but a very important language in which operation codes and operands are given in the form of alphanumeric symbols instead of 0's and 1's. These alphanumeric symbols will be known as mnemonic codes and can have maximum up to 5 letter combination e.g. ADD for addition, SUB for subtraction, START, LABEL etc. Because of this feature it is also known as „Symbolic Programming Language“. This language is also very difficult and needs a lot of practice to master it because very small

English support is given to this language. The language mainly helps in compiler orientations. The instructions of the Assembly language will also be converted to machine codes by language translator to be executed by the computer.

1	entry	main, ^m<r2>
2	subl2	#12, sp
3	jsb	C\$MAIN_ARGS
4	movab	\$CHAR_STRING_CON
5		
6	pushal	-8(fp)
7	pushal	(r2)
8	calls	#2, SCANF
9	pushal	-12(fp)
10	pushal	3(r2)
11	calls	#2, SCANF
12	mull3	-8(fp), -12(fp), -
13	pusha	6(r2)
14	calls	#2, PRINTF
15	clrl	r0
16	ret	

Advantages of Assembly Language

- i) It is easier to understand and use as compared to machine language.
- ii) It is easy to locate and correct errors.
- iii) It is modified easily

Disadvantages of Assembly Language

- i) Like machine language it is also machine dependent.
- ii) Since it is machine dependent therefore programmer Should have the knowledge of the hardware also.

High Level Languages

High level computer languages give formats close to English language and the purpose of developing high level languages is to enable people to write programs easily and in their own native language environment (English). High-level languages are basically symbolic languages that use English words and/or mathematical symbols rather than mnemonic codes. Each instruction in the high level language is translated into many machine language instructions thus showing one-to-many translation

Types of High Level Languages

Many languages have been developed for achieving different variety of tasks, some are fairly specialized others are quite general purpose.

These are categorized according to their use as

a) Algebraic Formula-Type Processing. These languages are oriented towards the computational procedures for solving mathematical and statistical problem

Examples are

- **BASIC (Beginners All Purpose Symbolic Instruction Code).**
- **FORTTRAN (Formula Translation).**
- **PL/I (Programming Language, Version 1).**
- **ALGOL (Algorithmic Language).**

- **APL (A Programming Language).**

b) Business Data Processing:

- These languages emphasize their capabilities for maintaining data processing procedures and files handling problems. Examples are:
- **COBOL (Common Business Oriented Language).**
- **RPG (Report Program Generator)**

b) String and List Processing: These are used for string manipulation including search for patterns, inserting and deleting characters. Examples are:

- **LISP (List Processing).**
- **Prolog (Program in Logic).**

Object Oriented Programming Language

In OOP, the computer program is divided into objects. Examples are:

- **C++**
- **Java**

e) Visual programming language: these are designed for building Windows-based applications Examples are:

- **Visual Basic**
- **Visual Java**
- **Visual C**

```

1  /* This program reads two integers from the keyboard
2     and prints their product.
3     Written by:
4     Date:
5  */
6  #include <stdio.h>
7
8  int main (void)
9  {
10 // Local Definitions
11     int number1;
12     int number2;
13     int result;
14
15 // Statements
16     scanf ("%d", &number1);
17
18     scanf ("%d", &number2);
19     result = number1 * number2;
20     printf ("%d", result);
21     return 0;
22 } // main

```

Advantages of High Level Language

Following are the advantages of a high level language:

- User-friendly
- Similar to English with vocabulary of words and symbols
- Therefore it is easier to learn.
- They are easier to maintain.

Disadvantages of High Level Language

- A high-level language has to be translated into the machine language by a translator and thus a price in computer time is paid.
- The object code generated by a translator might be inefficient Compared to an equivalent assembly language program

Creating and Running Programs:

There are four steps in this process.

1. Writing and editing the program using Text editor (source code).
2. Compile the program using any C compiler.(.bak file)
3. Linking the program with the required library modules(object file)
4. Executing the program. (.Exe file)

Creating and Editing a C Program in C Programming Language compiler:

Writing or creating and editing source program is a first step in c language. Source code is written in c programming language according to the type of problem or requirement, in any text editor.

Saving C Program in C Programming Language: Source code is saved on the secondary storage. Source code is saved as text file. The extension of file must be ".c". Example the file name is "learn c programming language.c"

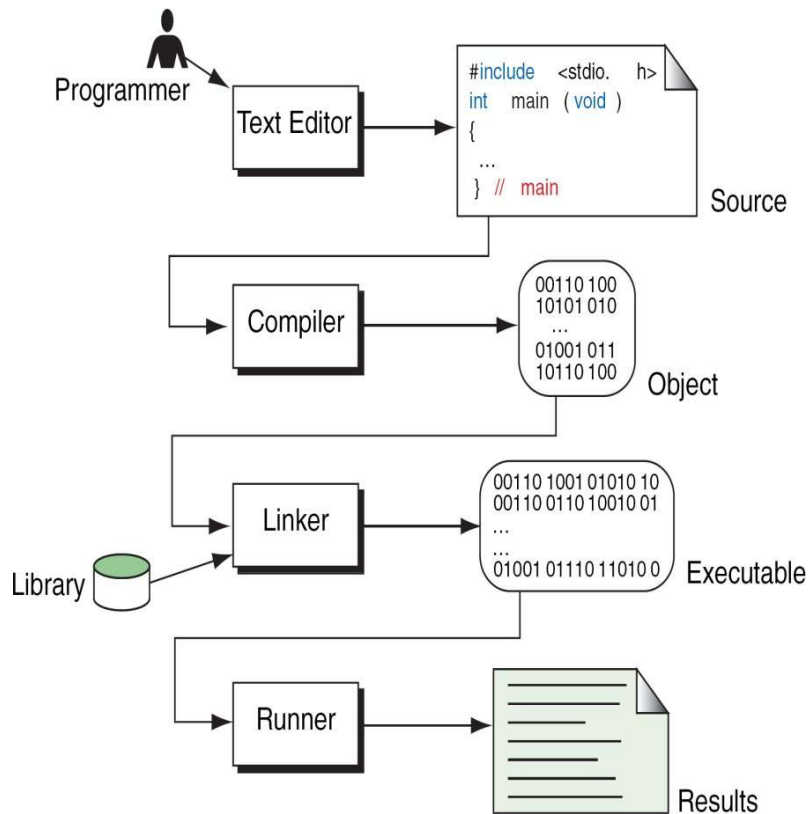
Compiling C program in C Programming Language: Computer does not understand c programming language. It understands only 0 and 1 means machine language. So c programming language code is converted into machine language. The process of converting source code in to machine code is called compiling. Compiler is a program that compiles source code. Compiler also detects errors in source program. If compiling is successful source program is converted into object program. Object program is saved on disk. The extension of file is ".obj"

Linking in C programming Language: There are many built in functions available in c programming language. These functions are also called library functions. These functions are stored in different header files.

Loading program: The process of transferring a program from secondary storage to main memory for execution is called loading a program. A program called loader

does loading.

Executing program: Execution is the last step. In this step program starts execution. Its instructions start working and output of the program display on the screen.



Pseudocode: is an artificial and informal language that helps programmers develop algorithms. Pseudocode is very similar to everyday English.

Algorithm:

An algorithm is a description of a procedure which terminates with a result. Algorithm is a step-by-step method of solving a problem.

Properties of an Algorithm:

- 1) **Finiteness:** - An algorithm terminates after a finite numbers of steps.
- 2) **Definiteness:** - Each step in algorithm is unambiguous. This means that the action specified by the step cannot be interpreted (explain the meaning of) in multiple ways & can be performed without any confusion.
- 3) **Input:** - An algorithm accepts zero or more inputs
- 4) **Output:-** An algorithm should produce at least one output.

5) Effectiveness: - It consists of basic instructions that are realizable. This means that the instructions can be performed by using the given inputs in a finite amount of time.

Writing an algorithm

An algorithm can be written in English, like sentences and using mathematical formulas. Sometimes algorithm written in English like language is Pseudo code.

Examples

- 1) Finding the average of three numbers
1. Let a,b,c are three integers
2. Let d is float
3. Display the message “Enter any three integers:”
4. Read three integers and stores in a,b,c
5. Compute the $d = (a+b+c)/3.0$
6. Display “The avg is:” , d
7. End.

- **Example 1:** Write an algorithm to determine a student’s final grade and indicate whether it is passing or failing. The final grade is calculated as the average of four marks.

Pseudocode::

- **Input a set of 4 marks**
- **Calculate their average by summing and dividing by 4**
- **if average is below 50**
 Print “FAIL”
 else
 Print “PASS”

■ **Detailed Algorithm :**

- **Step 1: Input M1,M2,M3,M4**
- Step 2: GRADE $\leftarrow (M1+M2+M3+M4)/4$**
- Step 3: if (GRADE < 50) then**
 Print “FAIL”
 else
 Print “PASS”
 endif

Flowcharts :

The pictorial representation of algorithm is called flowchart.

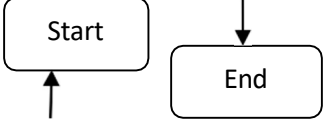
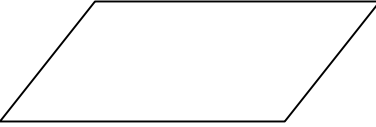
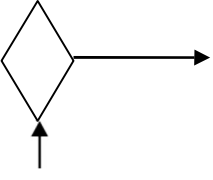
Uses of flow chart:

1 : flow chart helps to understand the program easily.

2 : as different symbols are used to specify the type of operation performed, it is easier to

understand the complex programs with the help of flowcharts.

Flowchart Symbols

S.NO	Description	Symbols
1	Flowlines : These are the left to right or top to bottom lines connection symbols. These lines shows the flow of control through the program.	
2	Terminal Symbol : The oval shaped symbol always begins and ends the flowchart. Every flow chart starting and ending symbol is terminal symbol.	
3	Input / Output symbol : The parallelogram is used for both input (Read) and Output (Write) is called I/O symbol. This symbol is used to denote any function of an I/O device in the program.	
4	Process Symbol : The rectangle symbol is called process symbol. It is used for calculations and initialization of memory locations.	
5	Decision symbol : The diamond shaped symbol is called decision symbol. This box is used for decision making. There will be always two exits from a decision symbol one is labeled YES and other labeled NO.	
6	Connectors : The connector symbol is represented by a circle. Whenever a complex flowchart is morethan one page, in such a situation, the connector symbols are used to connect the flowchart.	

Algorithm to find whether a number even or odd:

Step1: Begin

Step2: Take a number

**Step3: if the number is divisible by2 then
print that number is even
otherwise print that number is odd**

Step1: START

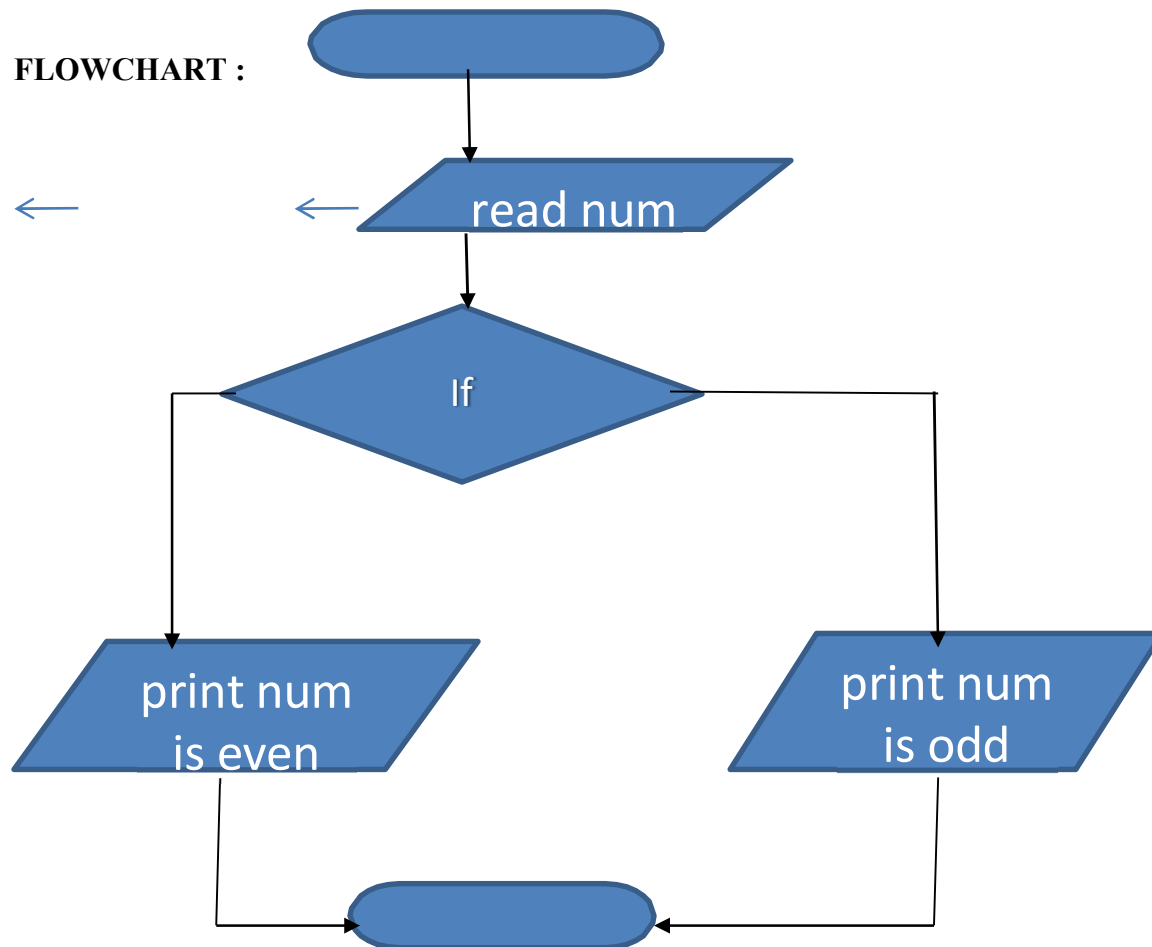
Step2: Read num

**Step3: if(num%2=0) then
print num is even
otherwise**

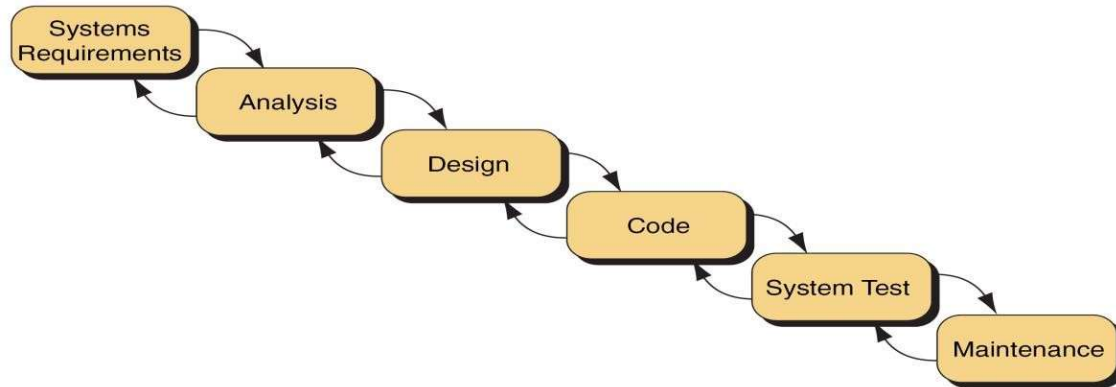
Step4: End
(Algorithm in natural language)

print num is odd
Step4: STOP
(Algorithm by using pseudo code)

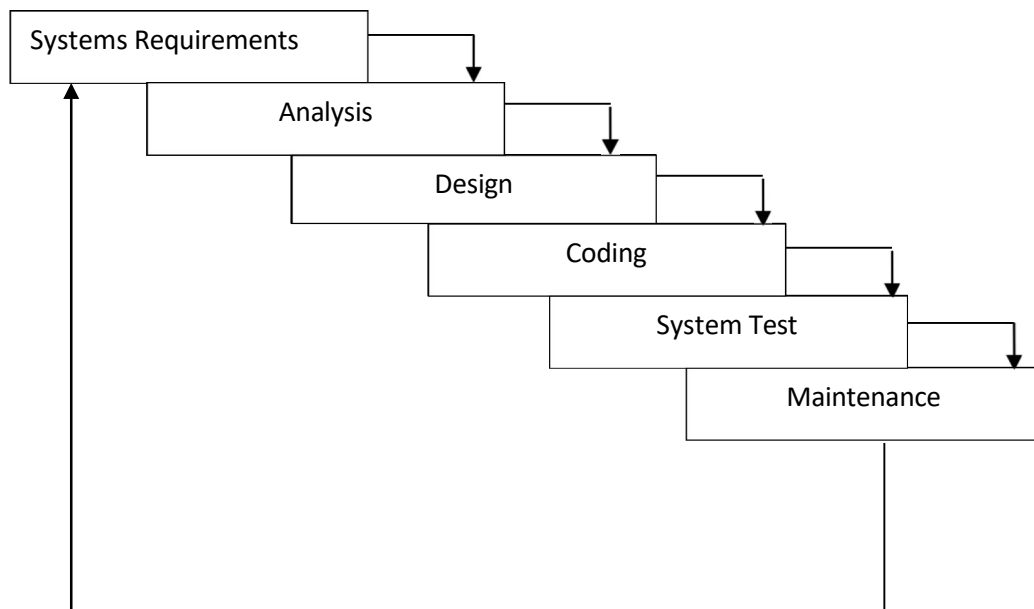
FLOWCHART :



System Development:



Or



1. Statement of Problem

- a) Working with existing system and using proper questionnaire, the problem should be explained clearly.
- b) What inputs are available, what outputs are required and what is needed for creating workable solution, should be understood clearly.

2. Analysis

- a) The method of solutions to solve the problem can be identified.
- b) We also judge that which method gives best results among different methods of solution.

3. Design

- a) Algorithms and flow charts will be prepared.
- b) Focus on data, architecture, user interfaces and program components.

4. System Test

The algorithms and flow charts developed in the previous steps are converted into actual programs in the high level languages like C.

a. Compilation

The process of translating the program into machine code is called as Compilation. Syntactic errors are found quickly at the time of compiling the program. These errors occur due to the usage of wrong syntaxes for the statements.

Eg: $x=a*y+b$

There is a syntax error in this statement, since, each and every statement in C language ends with a semicolon (;).

b. Execution

The next step is Program execution. In this phase, we may encounter two types of errors. Runtime Errors: these errors occur during the execution of the program and terminate the program abnormally.

Logical Errors: these errors occur due to incorrect usage of the instructions in the program. These errors are neither detected during compilation or execution nor cause any stoppage to the program execution but produces incorrect output.

5. Maintenance

We are maintenance the software by updating the information, providing the security and license for the software.

What is C?

C is a programming language developed at **AT & T's Bell Laboratories of USA in 1972**. It was designed and written by **Dennis Ritchie**. **Dennis Ritchie** is known as the **founder of c language**.

It was developed to overcome the problems of previous languages such as B, BCPL etc.

Initially, C language was developed to be used in UNIX operating system.

Features of C

1. Portability or machine independent